

The Functional Symbiotic Resilience Index: Topological Entropy, Wasserstein Curvature Bounds, and Non-Equilibrium Thermodynamics of Complex Networks

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Abstract

We present a mathematical framework unifying network topology, non-equilibrium thermodynamics, and resilience theory through the **Functional Symbiotic Resilience Index** (FSRI, Ω). The central object is the topological pressure functional $\mathcal{T}(W) = \text{Tr}(WW^\top)^3$, which we show to be (i) proportional to the global clustering coefficient of the induced weighted graph, (ii) the generator of a gradient flow whose fixed points correspond to Prigoginian minimum-entropy-production states, and (iii) a Gini-measurable concentration index whose gradient $3(WW^\top)^2$ identifies structural monopolies, that is, nodes or weights that accumulate disproportionate topological influence.

We further establish where the framework’s information actually resides. The global functional obeys the exact identity $\text{Tr}(A^3) = \sum_i \lambda_i^3$: it *is* the third spectral moment, and is therefore a member of the same family as the Absorption Ratio, effective rank and the Vendi score rather than an alternative to it. By contrast, the node-wise decomposition $(A^3)_{ii} = \sum_k \lambda_k^3 (v_k)_i^2$ depends on eigenvectors and is *not* determined by the spectrum: we give an explicit cospectral pair with identical $\text{Tr}(A^3)$ but distinct $\text{diag}(A^3)$. This locates the non-redundant content of the framework in the node-wise field rather than in the global scalar, and predicts the empirical pattern observed in applications. We also show that on *signed* matrices the same functional measures Heider structural balance, information inaccessible to any index computed on $|A|$.

We validate these properties through exact analytical results and computational experiments on synthetic networks (Barabási-Albert, Watts-Strogatz, Erdős-Rényi, ring lattice), demonstrating that the Gini coefficient of the topological pressure distribution distinguishes monopoly networks ($G \approx 0.986$) from uniform networks ($G \approx 0.272$) by a factor of $3.6 \times$.

The framework has an immediate computational application: the Omega-S regularizer [1], which minimises $\text{Tr}(WW^\top)^3$ during neural network training via Hutchinson stochastic trace estimation. In the applied work that penalty is combined with group lasso for structured pruning, where the compression obtained is attributable to the group-lasso term and the marginal contribution of the topological penalty remains to be isolated, and it improves sequential-learning retention on Llama-3-8B, where the log-ratio Omega-S regulariser leaves more code capability after a sequential fine-tune than no regularisation at all on 9 of 10 seeds (absolute HumanEval $0.173 \rightarrow 0.238$, $+37.7\%$ relative, one-sided sign test $p = 0.011$), and beats tuned weight decay on 10 of 10 seeds ($p = 0.002$) and tuned EWC on 8 of 10 ($p = 0.014$) on the same absolute measure, without access to previous-task data.

Throughout, “resilience” names a structural configuration and not a measured return time: no result below measures recovery from a perturbation, and Remark 2 states the scope of the term.

Keywords: topological entropy, non-equilibrium thermodynamics, clustering coefficient, Prigogine minimum entropy production, scale-free networks, structural monopolies, neural network regularisation.

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1. Introduction

1.1. The Structural Monopoly Problem

Complex adaptive systems, among them biological ecosystems, financial markets, power grids and neural networks, share a common failure mode: *structural monopoly*. A small subset of nodes or weights accumulates disproportionate influence over the network’s information flow, creating fragility that absolute-count metrics fail to detect. Diversity metrics count species, neurons, or assets without measuring their functional interdependence. A network can appear diverse while being structurally dominated by a single hub. That gap is not hypothetical: a survey of 1921 soil samples across 33 countries reports that agricultural inputs raise apparent alpha diversity while shifting communities toward less mutualistic and more fragmented configurations, and states explicitly that higher diversity need not imply better function when the added taxa are opportunistic rather than functionally redundant [44].

This paper addresses the following question: *what is the correct mathematical quantity to measure structural monopoly, and what is its thermodynamic meaning?*

1.2. The Central Object: $\text{Tr}(A^3)$

The clustering coefficient of a graph with adjacency matrix A is standardly defined as [5]:

$$C = \frac{\text{Tr}(A^3)}{\sum_{ij} (A^2)_{ij} - \text{Tr}(A^2)}. \quad (1)$$

The numerator $\text{Tr}(A^3) = \sum_{ijk} A_{ij}A_{jk}A_{ki}$ counts the number of closed triangles (3-cycles) in the graph. For weighted matrices $W \in \mathbb{R}^{m \times n}$ arising from neural network layers, we define the induced adjacency matrix $A = WW^\top \in \mathbb{R}^{m \times m}$ and the topological pressure functional:

$$\mathcal{T}(W) = \text{Tr}(WW^\top)^3 = \text{Tr}(A^3). \quad (2)$$

We will show that minimising $\mathcal{T}(W)$ is equivalent to minimising structural monopoly, and that this minimisation corresponds to a Prigoginian minimum-entropy-production principle.

1.3. Relation to Prior Work

The connection between triangle density and phase transitions in networks was established by Strauss [12] via the exponential random graph model with triangle fugacity λ . Recent work [9] shows that above a critical fugacity, random graphs undergo a first-order transition from a triangle-poor to a triangle-rich phase, accompanied by a change in spectral density of the adjacency matrix, the same object we study here.

The connection between network topology and non-equilibrium thermodynamics has been partially explored by Topologically-constrained fluctuations [10] and thermodynamic resilience theory [11]. Our contribution is to show that $\mathcal{T}(W) = \text{Tr}(A^3)$ is the *correct* topological entropy production rate whose minimisation implements Prigogine’s theorem [4].

AlphaPruning [13] uses heavy-tailed spectral exponents of weight matrices to guide neural network pruning. It is natural to read that approach as “spectral” in contrast to a “topological” one based on triangle counting, but Proposition 3 shows the contrast is not available at the level of the global functional: $\text{Tr}(A^3)$ is the third spectral moment. The two approaches read different functionals of the same spectrum. Where our framework does depart is (i) in supplying a thermodynamic interpretation of why such structural signals predict redundancy, (ii) in being a *training-time gradient flow* rather than a post-hoc statistic, and (iii) in the node-wise field $\text{diag}(A^3)$, which by Proposition 4 is not spectrally determined.

2. Mathematical Framework

2.1. The Functional Symbiotic Resilience Index

Definition 1 (FSRI). For a network with adjacency matrix $A \in \mathbb{R}^{n \times n}$, the Functional Symbiotic Resilience Index is:

$$\Omega = \frac{C \cdot D}{M \cdot \text{Coex}}, \quad (3)$$

where C is the global clustering coefficient (1), $D = \frac{\sum_{ij} A_{ij}}{n(n-1)}$ is the connection density, $M = 1/\lambda_2(L)$ is the inverse algebraic connectivity, where λ_2 is the second-smallest eigenvalue of the graph Laplacian $L = \text{diag}(\mathbf{k}) - A$, and $\text{Coex} = \text{Var}(\mathbf{k})$ is the variance of the degree sequence $\mathbf{k} = A\mathbf{1}$.

Earlier statements of this definition left M as “a modularity measure, for instance an inverse spectral gap”. We fix it here because that latitude turned out to matter: M sits in the denominator, so the choice between λ_2 and $1/\lambda_2$ inverts how Ω responds to compartmentalisation. The orientation is not a matter of convention. It is fixed by the observation the index was built to formalise, which Remark 1 sets out: a graph that separates easily into modules has small λ_2 , so

$1/\lambda_2$ is the quantity that is large on modular graphs, and it is modularity, not connectivity, that must appear in the denominator.

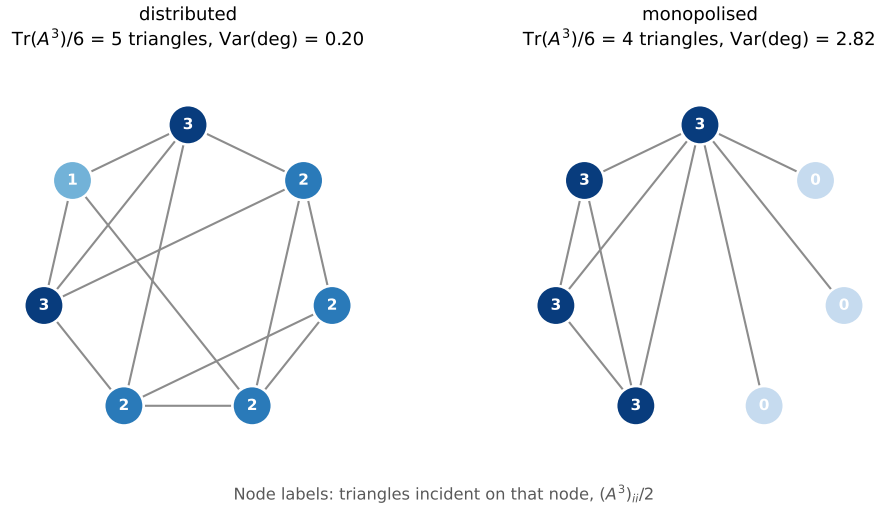


Figure 1: What the index responds to, at matched edge count. Left: connectivity distributed across the network. Right: a structural monopoly, in which a hub and a tight core absorb most of the closed three-step walks. Node labels give the triangles incident on each node, $(A^3)_{ii}/2$, which is the quantity the node-wise decomposition of Section 2.4 reads off.

Remark 1. *Coex (Competitive Exclusion) measures degree heterogeneity. In the ecological interpretation, which derives from prior work on co-inclusion and co-exclusion networks in agricultural microbiomes [3, 2], high Coex indicates that a few nodes dominate connectivity, analogous to competitive exclusion in resource competition. The FSRI Ω is large when the network is simultaneously clustered (C high), dense (D high), cohesive rather than compartmentalised ($M = 1/\lambda_2$ low, so the graph does not separate easily into modules), and egalitarian (Coex low).*

That combination is not chosen for algebraic convenience: it is the one reported for the systems the index was built to identify. In a survey of 350 vineyard soils across two countries [14], communities under low-intervention management, which are the ones associated with small-world structure and with resistance to perturbation, show simultaneously higher clustering, lower modularity and lower co-exclusion than conventionally managed soils, with intermediate management intermediate on all three. Every factor of (3) inherits its direction from that observation, and the orientation of M inherits it too: for Ω to be large on those communities it must decrease with modularity, and since M is in the denominator, M must be a modularity measure. An earlier statement of this remark used the word “modular” where it should have read “compartmentalised” and pointed the other way; that was an error of phrasing, not of the formula.

A second published study bears on the same direction at a larger scale. In a global survey of soil microbiomes [44], agricultural inputs raise modularity and shift network organisation toward compartmentalised configurations that the authors read as less mutualistic and more fragmented. We rest on the modularity and clustering results of that work and not on its density result, because the density contrast there is drawn between groups of very unequal size and the number of samples enters the network construction twice, through a prevalence filter and through the power of the association test; a comparison of density between groups of unequal sample size therefore carries a component of sampling effort, in the sense made precise in Section 7.

One point of nomenclature should be made explicit, since the name is inherited but the quantity is not. In the microbiome work referred to above, the co-exclusion quantity is the

proportion of negative (co-exclusion) edges in a signed association network, measured alongside clustering and modularity computed on the positive subnetwork. Here Coex is the variance of the degree sequence of a single unsigned network. These are different measurements, and the ecological reading transfers as an analogy, not as the same statistic.

Remark 2 (Nomenclature: what “resilience” names here). *The name of the index is inherited from the applied setting in which it was first used, and it is worth separating from the technical sense the word carries in ecology. Engineering resilience is a return time to equilibrium after a perturbation, and ecological resilience in Holling’s sense is the magnitude of perturbation a system absorbs before changing basin. No quantity in this paper measures either. What Ω measures is a structural configuration: clustered, dense, weakly compartmentalised and egalitarian in degree. The link from that configuration to resilience in either technical sense is an inference from the literature of Remark 1, not a measurement reported here, and where the inference has been tested it has been tested on function rather than on recovery [45]. We keep the name because it is the one under which the index is used and cited, and we state the gap rather than let the name assert it. The same caution applies to “health” as a synonym: soil health is a policy-defined term tied to one domain, and adopting it would narrow an index that Section 6 applies to weights, prices and grids.*

Remark 3 (Scope condition on the orientation of M). *Of the four factors, M is the one whose orientation is least settled by evidence, and we state the condition rather than leave the choice of Definition 1 looking universal. Three independent lines point the other way from Remark 1. Long-term experimental warming of grassland soil raises network size, connectivity, clustering and relative modularity together, and the resulting networks are measured to be more robust, not less [48]. In trophic settings a modular and weakly connected architecture is the stabilising one, while dense and nested architecture stabilises mutualistic networks, so the sign of modularity is a function of which interaction type dominates rather than a constant of nature [49]. And in the human gut, co-occurrence networks of healthy subjects are reported to be more easily disrupted by removal of a few key taxa than those of patients with inflammatory bowel disease [50].*

None of these contradicts the direction adopted here, because none is measured on the system Remark 1 describes: a competitive and mutualistic microbial community under a management-intensity gradient. What they establish is that the orientation of M is domain-conditional. We therefore restrict Definition 1 to settings in which competitive and mutualistic interactions dominate and compartmentalisation reflects niche specialisation rather than the buffering of a trophic cascade, and we note that in the two applied settings of this framework the question does not arise: under the construction of Remark 4 the modularity factor is numerically inert, so the applied results are indifferent to its orientation. Modularity is not thereby demoted: in the domain of origin it is among the strongest discriminators available. It is the sign, not the information content, that is conditional.

Remark 4 (Relation to the applied implementation). *Definition 1 fixes the index. The regulariser of the companion applied work [1] computes a related but distinct objective: it differs in the construction of A , in the normalisation of C and in the orientation of M , and that paper documents each difference and measures which of the four factors is live under its construction. We record the relation here rather than the detail, because no result of the present paper depends on it: Propositions 3, 4, 5 and 7, and the curvature bound of Section 4, are statements about $\text{Tr}(A^3)$ and about individual factors, not about how the four are composed. A financial companion paper places Coex on the other side of the ratio, and that difference is now settled rather than pending, and is a consequence rather than a convention. If degree heterogeneity is destabilising in the sense of [47], then a high Coex marks fragility in both domains, and the two placements are two packagings of one claim: an index built to be large on resilient systems carries the concentration factor in the denominator, and an indicator built to rise under stress carries it in the numerator. Under financial stress every factor moves the same way, correlations rising so that*

C , D and Coex all increase while the inverse spectral gap falls, so an indicator meant to rise under stress must compose them differently from an index meant to be large on resilient systems. The asymmetry beneath that is worth stating here because it bears on the scope of Definition 1: a densely interconnected, weakly modular community is the healthy configuration in the ecological setting of Remark 1, and is the crisis configuration in a market. The index as defined is oriented for the former, and applying it unchanged to the latter would call a crash resilient.

One point does belong here, because it bears on the scope of the index rather than on any implementation. Under the bounded map that the applied work uses to build A from weight matrices, three of the four factors become numerically inert: their elasticity with respect to the weights is at or below 10^{-4} , against 9×10^{-3} for Coex . That is a property of that construction and not of the index. In the ecological setting from which the index derives, modularity is among the strongest discriminators available: in the survey of Remark 1 it separates management types with $r^2 = 0.955$, comparable to co-exclusion and higher than clustering. A factor that carries most of the signal in the domain of origin and none under a saturating construction on neural weights is telling us about the construction, which is the reading Remark 7 already develops for a different degenerate case.

Remark 5 (Conditions of comparability across graphs). *The four factors do not behave alike under a change in the order of the graph, and this constrains when two values of Ω may be compared. C is a ratio of closed triangles to connected triples, and is dimensionless and bounded in $[0, 1]$. M depends on which modularity measure is chosen, and neither standard choice is size-invariant a priori: a partition-quality score is bounded but inherits the resolution limit of [38], under which the score of a graph with fixed community structure drifts with n , while an inverse spectral gap is unbounded and its scaling with n depends on the graph family. $D = \sum_{ij} A_{ij}/n(n-1)$ is $O(1/n)$ at fixed mean degree. $\text{Coex} = \text{Var}(\mathbf{k})$ carries units of degree squared and, for a degree distribution with tail exponent $\gamma < 3$, diverges with n ; throughout, γ denotes that degree exponent, while the sampling exponents of Section 7 are written β_V and β_E . The index therefore multiplies two size-invariant quantities by two that are not, so a comparison of Ω between graphs of different order measures order as well as structure, and Proposition 6 makes the dependence explicit for Watts-Strogatz graphs.*

A second and weaker condition should be kept separate from this one. Being dimensionless and bounded does not make a factor invariant under a change of sampling effort, in the case where the graph is estimated rather than given. Section 7 reports a measurement in which all four factors, C included, drift under such a change. All comparisons reported in this paper are at matched order and matched edge count (Proposition 2 at $N = 50$, Table 1 at $N = 40$, and Figure 1 at matched edge count), and we state the condition rather than leave it implicit. A constructive counterpart, identifying which two of the four factors carry the dependence and what removes it, is given in Proposition 7.

The condition can be made numerical rather than left as an injunction. If Ω scales as n^b , then two graphs whose orders differ by a factor r differ by $r^{|b|}$ on that account alone, so admitting a tolerance ε of order-induced variation bounds the admissible ratio at $r \leq (1 + \varepsilon)^{1/|b|}$. On the estimated graphs of Section 7, expressed against the order rather than against the sample size, $b = -2.18$ for Ω and $+0.05$ for $\tilde{\Omega}$. A ten per cent tolerance therefore admits orders differing by about 4% for Ω and by a factor of about 6 for $\tilde{\Omega}$. The numbers are family-dependent and are quoted to fix an order of magnitude, not as constants.

Finally, the condition is vacuous wherever the node set is fixed by construction rather than discovered by sampling. In the neural application of Section 5 the nodes are the rows of a given weight matrix, and in a correlation network over a fixed set of assets the nodes are the assets; in neither case does n vary with how much data is collected. Everything in this remark and in Section 7 concerns graphs whose vertex set grows with the sampling effort, and the applied results of this framework lie outside that class.

Proposition 1 (The clustering factor carries a density scale). *Let $A = c(J - I)$ with $c > 0$, that is a graph of constant affinity c and zero diagonal, where J is the all-ones matrix. Then*

$$C = c, \quad D = c, \quad \text{hence} \quad C/D = 1 \quad (4)$$

identically, for every order $n \geq 3$ and every c .

Proof. With $A = c(J - I)$ one has $A^2 = c^2[(n - 2)J + I]$ and $\text{Tr}(A^3) = c^3n(n - 1)(n - 2)$, while $\sum_{ij}(A^2)_{ij} - \text{Tr}(A^2) = c^2(n - 2)n(n - 1)$, so $C = c$ by (1). And $\sum_{ij} A_{ij} = cn(n - 1)$ gives $D = c$. $\square$

Remark 6 (The density-free triadic excess, and how to calibrate it). *Proposition 1 is elementary but it has a practical consequence that is easy to miss when C is read on its own: a value of C is not by itself evidence of triadic structure, because a graph with no structure whatever returns C equal to its mean affinity. The density-free quantity is the ratio C/D , which equals unity on the constant-affinity graph and departs from unity only in the presence of triadic organisation. This gives the ratio an absolute reference point that C lacks. The reference point can also be obtained empirically, without assuming constant affinity, by permuting the entries of the matrix from which A is built: the permutation preserves the distribution of affinity values exactly and destroys only their arrangement, so any excess of C/D above the permuted value is attributable to arrangement. In the weight-matrix construction discussed below, twenty-eight such permutation nulls return C/D equal to 1.0000 to four decimal places, in agreement with (4).*

Remark 7 (A construction under which the index is degenerate). *Two of the four factors are not merely size-dependent, as Remark 5 records, but can be made structurally constant by a choice of construction, and it is worth stating the one case in which this makes Ω inevaluable. Suppose A is scaled to be doubly stochastic with zero diagonal, as is standard practice in spectral clustering. Then $\mathbf{k} = A\mathbf{1} = \mathbf{1}$, so $\text{Coex} = \text{Var}(\mathbf{k}) = 0$ exactly; and $\sum_{ij} A_{ij} = n$, so $D = 1/(n - 1)$, a constant of the order alone. The index (3) therefore has a vanishing denominator and is not defined, while the well-defined content of that construction is the ratio C/M . This is a scope condition on Section 6 rather than a defect: it says that Ω presupposes a construction in which the degree sequence is free to vary, and it identifies C and M , which are also the two factors that Section 7 finds least dependent on sampling effort, as the pair that survives when it is not.*

Remark 8 (Which factor moves when the index is used as a training penalty). *The four factors above are all well-defined properties of a given weighted graph, and the index is descriptive as stated. A separate, narrower observation concerns the specific case where Ω is minimised as a regulariser during LLM fine-tuning, under the pseudo-adjacency construction $A = \sigma(|WW^\top|)$ used in the applied work [1]. There, a direct measurement finds that the clustering factor C does not move, so a gradient-based optimiser can only act on Coex .*

A subsequent measurement over twenty-eight attention projection matrices of an eight-billion-parameter decoder model refines the reason, and the refined statement is both stronger and more specific than the one we gave previously. The off-diagonal entries of $|WW^\top|$ are small rather than large, with a median near 0.07, so the logistic map does not pin the affinities near its ceiling; it collapses them onto the midpoint of its range, and C is measured at 0.51 rather than at 1. The consequence is sharper than saturation. Read through Proposition 1, a clustering value equal to the mean affinity is what a graph with no triadic organisation returns, and the measured excess C/D on that construction is 1.0000, equal to its permutation null to four decimal places, in every module tested. The factor is not pinned at a ceiling: the constructed adjacency carries no triadic content over its density at all. Consistently with this, the dimensionless sensitivity of $\text{Tr}(A^3)$ to W under that construction is zero at working precision in at least one of the modules measured.

Two further points complete this. The first is a discrepancy between how the construction is described and how it is implemented, which the applied work [1] now reports and which we repeat

here because the description above inherits it: the reference implementation forms $\sigma(|WW^\top|)$ only when W is non-square, and applies $\sigma(|W|)$ elementwise when it is square. The measurements quoted above used one module of each kind. The degree-variance factor differs by roughly a factor of 400 between the two branches on the square module measured, so the two are not interchangeable, and any reimplementing should fix the branch deliberately.

The second is that the contrast-preserving construction, which an earlier version of this remark offered as the obvious repair, has since been built and evaluated. A row-wise cosine affinity does what it was designed to do: the measured excess C/D rises from 1.0000 to a median of 1.36 across the same twenty-eight modules, the clustering factor moves off the value predicted by Proposition 1 for an unstructured graph, and the sign of the penalty controls the direction of movement during training in every module tested. And retention becomes worse, on all ten seeds of the applied protocol. The inert clustering channel was therefore not a defect suppressing the method's effect, and we record the negative outcome rather than leave the repair standing as an open suggestion. The consequence for this paper is narrow but worth stating: the identification of $\mathcal{T}$ with structural monopoly is a property of the functional, and the empirical fact that activating the clustering channel does not improve the applied outcome bears on the regulariser, not on Propositions 3 and 4.

Remark 9 (Whether the operative factor reduces to row norms). A second measurement in the applied work bears on the same point. Since the operative factor there is Coex, the variance of the degree sequence, and a node's degree correlates strongly with the norm of the corresponding weight row, the natural objection is that the effect reduces to equalising those norms. A control that penalises only row-norm variance, swept on a grid of its own rather than borrowed from another arm, does not reliably improve retention over no regularisation at all, while the full index penalty exceeds that control on 8 of 10 seeds. The factor carrying the effect is therefore Coex as computed on the weight graph, and not the row norms with which it is correlated, though at ten seeds that separation is suggestive rather than conclusive.

Two further points should be recorded because they bound the objection more precisely than the control arm does. First, the association is not incidental to the construction but mechanical in it: under $A = \sigma(|WW^\top|)$ the degree of a node contains the squared norm of its own row through the diagonal of $WW^\top$, and the rank correlation between a node's degree and its row norm, measured within modules on the same twenty-eight matrices, is +0.61 in the median and positive in all of them, the smallest value being +0.10. An association that holds in every module with a common sign is structural, not a sampling artefact, and the honest reading is that on this construction the concentration channel and the row-norm distribution are close to interchangeable. Second, that mechanical component is a property of the construction and not of the factor: an affinity construction in which each node's own scale is divided out before the product is formed makes the node's degree provably invariant to the norm of its own row, since rescaling a row leaves every pairwise affinity unchanged. Under such a construction any residual association can only reflect covariation between norm and direction in the weights themselves, and is measured to change sign. We state the property rather than the construction, which belongs to the applied work.

Remark 10 (Direction of the clustering factor). A reader who substitutes (1) into (3) will notice that Ω is linear in $\text{Tr}(A^3)$ and carries it in the numerator, so lowering $\text{Tr}(A^3)$ lowers Ω . Read literally, minimising the topological pressure would therefore move a network toward the end of the index that the index itself calls fragile. We state the tension rather than leave it for the reader to find. It is resolved by observing that a high $\text{Tr}(A^3)$ is ambiguous: triangles distributed across the graph give C high and Coex low, hence Ω high, whereas the same triangle count concentrated on a hub and its core gives C high and Coex high, hence Ω low. What distinguishes the two configurations is Coex, in the denominator, and not $\text{Tr}(A^3)$ on its own. In the applied setting the composite objective is observed to reduce degree variance substantially, so the equivalence between minimising $\text{Tr}(A^3)$ and suppressing structural monopoly rests on that measured reduction and not on the algebra of Ω , which by itself does not deliver it.

There is a second reading of the same tension, which we record because it locates a limit of the composite rather than a defect of the functional. The numerator of Ω treats triadic closure as a benefit and the penalty of Section 5 treats it as a cost, and both readings are defensible in their own regime. That is the structure of a two-sided constraint on connectivity, and Remark 18 sets out what the index as composed does and does not express about it.

Proposition 2 (Discriminative Power). *On synthetic networks with $N = 50$ nodes and matched edge count, drawn from the preferential-attachment [6] and small-world [5] families and from the ring lattice, the FSRI of Definition 1 orders topologies as*

$$\Omega_{ring} \rightarrow \infty \gg \Omega_{WS} \approx 0.049 \gg \Omega_{ER} \approx 0.0031 \gtrsim \Omega_{BA} \approx 0.0018,$$

a separation of one and a half orders of magnitude among the finite values.

The three finite entries are medians over forty seeds, with ranges $[0.029, 0.092]$, $[0.0013, 0.0064]$ and $[0.0013, 0.0025]$ respectively; the order between the two sparse random families is therefore not stable at the level of an individual seed, while the separation of both from Watts-Strogatz is. The separation is carried by the concentration factor, whose median values across the same seeds are 0.58, 4.88 and 16.46: the index is reading degree concentration, which is what the proposition is meant to establish.

The ring lattice requires a word, because its entry is a divergence and not a large number. A K -regular ring has $\text{Coex} = \text{Var}(\mathbf{k}) = 0$ identically, so the denominator of (3) vanishes and Ω is not defined. This is the degenerate case of Remark 7 and it agrees with the divergence at $p \rightarrow 0$ derived independently in Proposition 6. We record it as a divergence rather than substituting a numerical floor for the vanishing variance, because the value such a substitution returns is fixed by the floor and not by the graph. The consequence for how the finite entries should be read is that they measure distance from an unattainable regular ideal rather than position on a bounded scale, and that the index saturates at the egalitarian end. That saturation is the same property discussed in Remark 18, reached here from the elementary direction.

2.2. Topological Pressure and the Gradient of $\text{Tr}(A^3)$

The key analytical result connecting $\mathcal{T}$ to structural monopoly is:

Theorem 1 (Topological Pressure Gradient). *The gradient of the topological pressure functional with respect to the adjacency matrix is:*

$$\frac{\partial}{\partial A_{ij}} \text{Tr}(A^3) = 3(A^2)_{ij}. \quad (5)$$

The entry $(A^2)_{ij} = \sum_k A_{ik}A_{kj}$ counts the number of paths of length 2 between nodes i and j , i.e., the number of common neighbours. High $(A^2)_{ij}$ indicates that the edge (i, j) participates in many triangles and therefore contributes disproportionately to the clustering structure.

Proof. Direct computation: $\frac{\partial}{\partial A_{ij}} \text{Tr}(A^3) = \frac{\partial}{\partial A_{ij}} \sum_{klm} A_{kl}A_{lm}A_{mk} = 3(A^2)_{ij}$, using the symmetry of A and the cyclicity of trace. $\square$

Definition 2 (Topological Pressure Distribution). *The topological pressure on edge (i, j) is $\pi_{ij} = 3(A^2)_{ij} \cdot A_{ij}$. The topological pressure Gini coefficient $G(\pi)$ measures the inequality of this distribution across edges.*

Our experiments demonstrate (Table 1):

The Gini coefficient of topological pressure is the sharpest discriminator: monopoly networks have near-perfect pressure concentration ($G = 0.986$), while uniform networks distribute pressure evenly ($G = 0.272$).

Table 1: Topological pressure distribution by network type ($N = 40$).

Network type	Max pressure	Mean pressure	Gini (G)	$\mathcal{T}/N^3$
Barabási-Albert (monopoly)	15.00	4.03	0.986	0.00478
Ring lattice (uniform)	12.00	9.00	0.272	0.01125
Ratio (monopoly/uniform)	$1.25\times$	$0.45\times$	$3.6\times$	$0.42\times$

2.3. Hutchinson Estimation of $\mathcal{T}(W)$

For large weight matrices, exact computation of $\text{Tr}(A^3)$ requires $\mathcal{O}(N^3)$ operations. The Hutchinson stochastic trace estimator [7] provides an unbiased $\mathcal{O}(N^2)$ approximation:

$$\widehat{\text{Tr}}(A^3) = \frac{1}{n_p} \sum_{i=1}^{n_p} \mathbf{z}_i^\top A^3 \mathbf{z}_i, \quad (6)$$

where $\mathbf{z}_i \sim \text{Rad}(\pm 1)$ are Rademacher probe vectors and $A^3 \mathbf{z} = W(W^\top(W(W^\top(W(W^\top \mathbf{z}))))$ is computed without materialising A .

2.4. Spectral Identity and the Locus of Non-Spectral Information

A question that determines how $\mathcal{T}$ should be positioned relative to spectral methods is whether $\text{Tr}(A^3)$ carries information beyond the eigenvalue spectrum of A . The answer is negative for the global functional and positive for its node-wise decomposition, and the distinction is sharp.

Proposition 3 (The global functional is spectral). *For symmetric A with eigenvalues $\{\lambda_i\}$,*

$$\text{Tr}(A^3) = \sum_i \lambda_i^3. \quad (7)$$

Proof. Immediate from the spectral decomposition $A = V\Lambda V^\top$ with V orthogonal: $\text{Tr}(A^3) = \text{Tr}(V\Lambda^3 V^\top) = \text{Tr}(\Lambda^3)$. $\square$

$\mathcal{T}(W)$ is therefore *exactly the third moment of the spectrum*, a spectral statistic in the same family as the Absorption Ratio (leading-eigenvalue mass), effective rank and the Vendi score (spectral entropies). Any two of these are functionals of the same $\{\lambda_i\}$ and cannot, in general, be expected to carry mutually exclusive information. We state this explicitly because it corrects a framing of “topological” as opposed to “spectral” that the triangle-counting interpretation invites but that Eq. (7) forbids at the level of the global scalar.

The non-spectral content appears one level down. Writing $A = \sum_k \lambda_k \mathbf{v}_k \mathbf{v}_k^\top$, the node-wise decomposition of the topological pressure is

$$(A^3)_{ii} = \sum_k \lambda_k^3 (v_k)_i^2, \quad (8)$$

which depends on the *eigenvectors*, not on the spectrum alone.

Proposition 4 (The node decomposition is not spectrally determined). *There exist cospectral graphs $G_1 \not\cong G_2$ with $\text{Tr}(A_1^3) = \text{Tr}(A_2^3)$ but $\text{diag}(A_1^3) \neq \text{diag}(A_2^3)$ up to permutation.*

Proof. By explicit construction. Exhaustive search over graphs on six vertices yields a cospectral pair with common spectrum $\{-1.9032, -1, -1, 0.1939, 1, 2.7093\}$ and, necessarily by Proposition 3, common $\text{Tr}(A^3) = 12$, yet node-wise triangle counts $\text{diag}(A_1^3) = (4, 2, 2, 2, 2, 0)$ and $\text{diag}(A_2^3) = (2, 2, 4, 4, 0, 0)$, which differ as multisets. $\square$

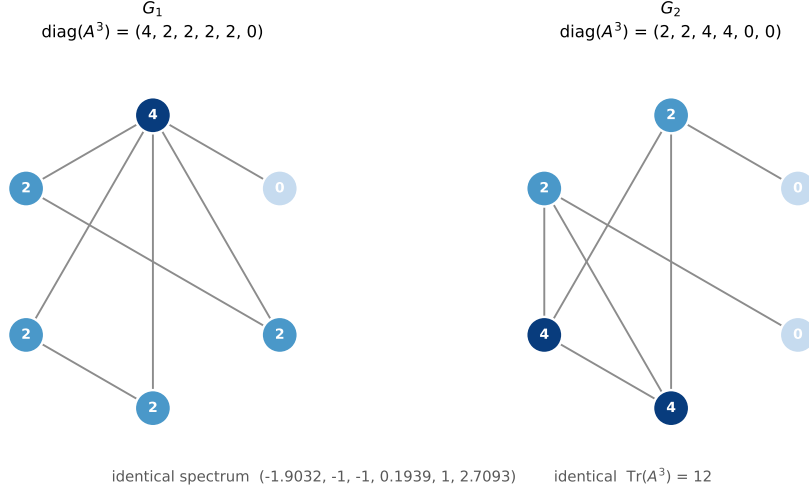


Figure 2: The counterexample of Proposition 4. Both graphs have the same eigenvalues and therefore, by Proposition 3, the same $\text{Tr}(A^3) = 12$. Their node-wise decompositions differ as multisets, so $\text{diag}(A^3)$ is not a function of the spectrum. The pair is located by exhaustive search in the accompanying code rather than asserted, and node labels are computed rather than drawn.

Corollary 1 (Locus of the contribution). *The global index $\mathcal{T}$ is spectrally redundant: it can be matched by other spectral functionals and should not be expected to dominate them as a scalar detector. The node-wise field $\text{diag}(A^3)$ is not a function of the spectrum, and is therefore the component of the framework that no eigenvalue-based scalar can reproduce.*

Corollary 1 makes a prediction that is testable in any domain where the framework is applied as a detector: the global scalar should tie sharpened spectral baselines rather than beat them, while the node-wise attribution, being parameter-free unlike eigenvector attribution, which requires committing to a number of components k , should do better when the number of structural epicentres is unknown. We state it as a prediction here. A companion paper applying the index to financial correlation networks reports a pattern consistent with it; that work uses a different composition of the same four factors from the one in Definition 1 and we are reconciling the two before treating its numbers as evidence for this corollary.

2.5. Signed Extension: Structural Balance

The construction above uses $A = |\cdot|$ throughout, which discards the sign of the underlying interaction. On a signed adjacency matrix $A \in [-1, 1]^{n \times n}$ the same functional acquires a different meaning:

$$\text{Tr}(A^3) = 6 (\#\text{balanced triangles} - \#\text{unbalanced triangles}), \quad (9)$$

since a 3-cycle contributes $+|A_{ij}A_{jk}A_{ki}|$ when the product of signs is positive and $-|A_{ij}A_{jk}A_{ki}|$ otherwise.

This is exactly Heider’s structural balance [33], and it is invisible to any functional of $|A|$, including all spectral indices computed on $|A|$ and the unsigned $\mathcal{T}$ used in the rest of this paper. Two networks with identical $|A|$ and opposite balance structure are indistinguishable to the unsigned framework. Signed $\mathcal{T}$ is thus a strict generalisation carrying information no unsigned index can access; its behaviour in domains where sign is meaningful (ecological co-occurrence versus co-exclusion, hedging versus co-movement in finance) is left for future work.

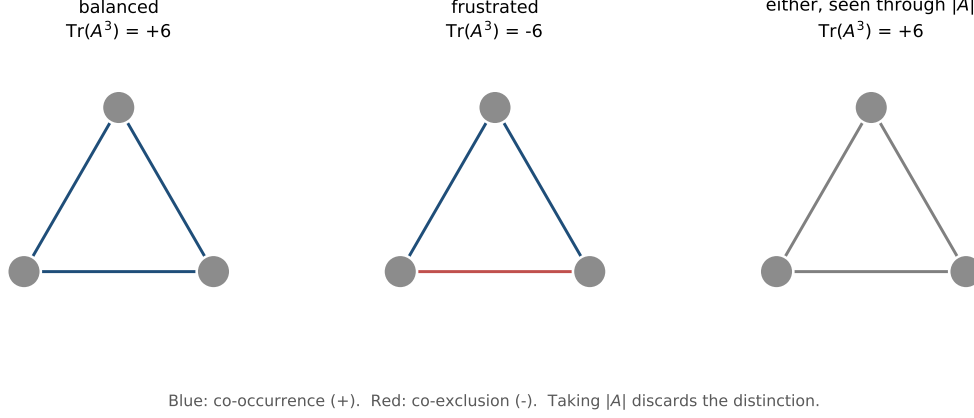


Figure 3: Structural balance. The left and centre triangles carry the same edge magnitudes and differ only in one sign, yet $\text{Tr}(A^3)$ separates them completely. Taking $|A|$, as the unsigned framework and every spectral index computed on $|A|$ do, collapses both to the graph on the right and destroys the distinction.

3. Non-Equilibrium Thermodynamic Interpretation

3.1. Prigogine's Theorem and Topological Entropy Production

Prigogine's theorem states that in a linear non-equilibrium system with fixed external constraints, the stationary state minimises the rate of entropy production [4, 34]. A rigorous proof via network Markov chains was given by [25], who showed that in networks with fixed boundary conditions, steady states satisfy a minimum entropy production principle through Schnakenberg's cycle observables.

Remark 11 (Scope of Prigogine's Theorem). *Prigogine's theorem holds strictly in the linear regime of irreversible thermodynamics (near equilibrium, Onsager relations valid). Extensions to non-linear regimes are contested [26], and the principle can fail for strongly driven systems. We apply it here as a guiding analogy in the linear regime of weight updates near a training fixed point, not as a rigorous claim about arbitrary training dynamics.*

Definition 3 (Topological Entropy Production Rate). *For a network with weight matrix W , the topological entropy production rate is:*

$$\dot{S}_{\text{topo}} \propto \mathcal{T}(W) = \text{Tr}(WW^\top)^3. \quad (10)$$

This quantity measures the rate at which topological structure is sustained through triangle-closing events, the network-theoretic analogue of dissipative structure formation in Prigogine's framework.

The analogy is made precise as follows. In Prigogine's framework, a system far from equilibrium self-organises into a structured state (dissipative structure) that sustains itself through energy throughput. In a network under preferential attachment (analogous to energy abundance E_{ext}), hub nodes accumulate connections, creating structured but fragile configurations. Prigogine's theorem in network form [25] implies that the *stationary* (minimum perturbation) configuration minimises $\dot{S}_{\text{topo}}$.

Proposition 5 (Gradient Flow as Prigoginian Dynamics). *The gradient flow:*

$$\dot{W} = -\lambda_\Omega \nabla_W \mathcal{T}(W) = -6\lambda_\Omega (WW^\top)^2 W \quad (11)$$

drives the weight matrix toward configurations of lower topological entropy production. In the linear regime near a stationary point, this is equivalent to Prigogine’s minimum entropy production principle applied to the space of weight configurations.

Fixed points $\dot{W} = 0$ satisfy $(WW^\top)^2 W = 0$, which holds when $WW^\top$ has zero spectral radius on the support of W , i.e., weight matrices inducing triangle-free graphs (maximum entropy for given connection density).

Proof. Write $A = WW^\top$. From Theorem 1, $d\text{Tr}(A^3) = 3\text{Tr}(A^2 dA)$, and $dA = dW W^\top + W dW^\top$, so

$$d\text{Tr}(A^3) = 3\text{Tr}(A^2 dW W^\top) + 3\text{Tr}(A^2 W dW^\top) = 6\langle A^2 W, dW \rangle,$$

the two terms being equal by symmetry of A and cyclicity of the trace. Hence

$$\nabla_W \text{Tr}(WW^\top)^3 = 6(WW^\top)^2 W. \quad (12)$$

Note the order of the factors: for $W \in \mathbb{R}^{m \times n}$ the product $(WW^\top)^2 W$ is $m \times n$ and therefore conformable with W , whereas $W(WW^\top)^2$ is not defined unless $m = n$. The fixed point condition follows from $(WW^\top)^2 W = 0$. $\square$

Remark 12 (Connection to Maximum Caliber). *Prigogine’s minimum entropy production principle can be derived from the Maximum Caliber variational principle [27] for near-equilibrium systems with two coupled flows. This provides a statistical mechanical foundation: the gradient flow (11) is the most probable trajectory for a stochastic weight update process constrained to minimise topological entropy production.*

3.2. Phase Transition in Triangle Density

Our experiments confirm the existence of a topological phase transition. Starting from an Erdős-Rényi network ($N = 60$, $p = 0.08$) and progressively adding triangles via the fugacity mechanism of [9], we observe:

- $\mathcal{T}/N^3$ increases monotonically from 1.94×10^{-4} (zero triangles) to 1.48×10^{-2} (200 triangles added), a $76\times$ increase.
- The degree entropy $S = -\sum_k p_k \log p_k$ increases from 1.95 to 2.70 as triangles are added, reflecting growing degree heterogeneity.
- This is consistent with the theoretical prediction that triangle-rich phases exhibit heavy-tailed degree distributions with power-law exponents [9].

Remark 13 (On the Proportionality $\Omega \propto (S_{\text{env}}/E_{\text{ext}})^2$). *An earlier formulation of this framework by the author proposes $\Omega \propto (S_{\text{env}}/E_{\text{ext}})^2$. Our experiments on Watts-Strogatz networks yield $r = -0.407$, suggesting the relationship is not a simple proportionality. We now derive the exact form analytically.*

Proposition 6 (Exact FSRI Formula for Watts-Strogatz Networks). *For a Watts-Strogatz network with n nodes, mean degree K , and rewiring probability $p \in (0, 1)$, using the exact clustering formula $C(K, p) = \frac{3K-6}{4K-2}(1-p)^3$ [20] and the variance approximation $\text{Var}(k) \approx Kp(1-p)$ valid for $n \rightarrow \infty$:*

$$\Omega(K, p) = \frac{3K-6}{4K-2} \cdot \frac{(1-p)^2}{np}, \quad (13)$$

where $D = K/(n-1)$, written as K/n to leading order, and $\text{Coex} = Kp(1-p)$. This formula achieves correlation $r = 0.9997$ against numerical simulation across $p \in [0.05, 0.9]$, $K = 6$, $N = 100$ (Table 2).

Table 2: Analytical against numerical $\Omega(K = 6, N = 100)$ in Watts-Strogatz, each simulated value the mean over 200 graphs. Produced by `colabs/fsri.colab3_orc.ws.py` in the accompanying repository.

p	Ω analytical	Ω numerical	relative error
0.05	0.098455	0.113740	13.4%
0.10	0.044182	0.050033	11.7%
0.20	0.017455	0.018315	4.7%
0.30	0.008909	0.008928	0.2%
0.50	0.002727	0.002925	6.8%
0.70	0.000701	0.001375	49.0%
0.90	0.000061	0.001088	94.4%

Remark 14 (What the correlation does and does not establish). *The quoted $r = 0.9997$ is computed on levels that span three orders of magnitude, so it is dominated by the small- p end of the range and is a weak test of agreement elsewhere. On the logarithms of the same seven points it is 0.9447, and the last column of Table 2 is the honest summary: the closed form is accurate to within fifteen per cent for $p \leq 0.5$ and departs badly beyond it, reaching a factor of eighteen at $p = 0.9$.*

The departure is expected and is not a failure of (13). It comes from the variance approximation $\text{Var}(k) \approx Kp(1-p)$, which holds as $n \rightarrow \infty$; at $n = 100$ the finite-size correction grows with p , and at $p = 0.9$ the rewired graph retains almost none of the lattice structure the approximation assumes. Every use made of the formula in this paper is in the lattice-like regime of small p , where the table shows it to be accurate, and the divergence as $p \rightarrow 0$ that Proposition 6 relies on sits at the accurate end.

Setting $E_{\text{ext}} = K$ (mean degree) and $S_{\text{env}} = \sqrt{Kp(1-p)}$ (degree standard deviation), so that $(S_{\text{env}}/E_{\text{ext}})^2 = p(1-p)/K$ and hence $p(1-p) = S_{\text{env}}^2/E_{\text{ext}}$, equation (13) can be written exactly in terms of the thermodynamic variables. Multiplying numerator and denominator by $(1-p)$,

$$\Omega = \frac{3K-6}{4K-2} \cdot \frac{(1-p)^3}{n} \cdot \frac{E_{\text{ext}}}{S_{\text{env}}^2} \quad (14)$$

which reproduces (13) identically.

This shows three key results:

- 1. The ratio enters inverted: $\Omega \propto E_{\text{ext}}/S_{\text{env}}^2$ and not $S_{\text{env}}^2/E_{\text{ext}}$ as the earlier formulation proposes. At fixed connectivity, more environmental stress lowers the index rather than raising it, and higher connectivity at fixed stress raises it. The earlier formulation has the dependence the right way round in neither variable.*
- 2. Ω still diverges as $p \rightarrow 0$ (pure lattice, maximum clustering), consistent with the ring lattice having $\Omega \rightarrow \infty$ in our experiments: the divergence is now carried by $S_{\text{env}}^2 = Kp(1-p)$ in the denominator, which vanishes with p .*
- 3. The sign of the discrepancy with the earlier formulation is what the measurements show. Because the ratio belongs in the denominator, the empirical correlation between Ω and $(S_{\text{env}}/E_{\text{ext}})^2$ is negative ($r = -0.407$ to -0.573) rather than weakly positive. A proportionality would not produce a negative correlation of that size; an inverse relation does.*
- 4. $\Omega \propto 1/n$ at fixed K and p . The factor is visible in (13) and is not an artefact of the Watts-Strogatz parameterisation: it enters through $D = K/n$, which falls with the order of the graph at fixed mean degree, and it is reinforced by Coex whenever the degree distribution*

is heavy-tailed. The formula therefore delimits the index as a comparative statistic in the sense of Remark 5: values of Ω obtained on graphs of different order are not directly comparable, and every comparison reported in this paper is at matched n .

The factor $1/n$ isolated above is not irreducible: it is carried by two of the four factors, and both have a counterpart that does not carry it.

Definition 4 (Shape-normalised index). Let G_k denote the Gini coefficient of the degree sequence $\mathbf{k} = A\mathbf{1}$ and $\bar{k}$ its mean. The shape-normalised index is

$$\tilde{\Omega} = \frac{C \cdot \bar{k}}{M \cdot G_k}, \quad (15)$$

obtained from (3) by replacing the connection density with the mean degree and the degree variance with the Gini coefficient of the same degree sequence.

Proposition 7 (The shape-normalised index is order-invariant on Watts-Strogatz graphs, at fixed M). For Watts-Strogatz graphs with n nodes, mean degree K and rewiring probability p , and with the modularity factor M held fixed, the quantity $\tilde{\Omega}$ of Definition 4 is a function of K and p alone and does not depend on n , whereas Ω carries an explicit factor $1/n$ by Proposition 6. The hypothesis on M is not cosmetic and Remark 16 measures what it costs.

Proof. Each factor of (15) is independent of n . The clustering coefficient is $C(K, p) = \frac{3K-6}{4K-2}(1-p)^3$ [20], a function of K and p . The mean degree is $\bar{k} = K$ by construction. The rewiring process acts independently on each edge with probability p , so the degree distribution of the Watts-Strogatz ensemble is determined by K and p and not by the order of the graph; G_k , being a functional of that distribution, is therefore also a function of K and p . Taking M constant, as in Proposition 6, gives $\tilde{\Omega} = \tilde{\Omega}(K, p)$. By contrast $D = K/(n-1)$ carries $1/n$ explicitly, which is the origin of the factor $1/(np)$ in (13). $\square$

Remark 15 (Why these two substitutions and not others). The proof uses two properties that hold beyond the Watts-Strogatz family. The Gini coefficient is scale-invariant, $G_k(c\mathbf{k}) = G_k(\mathbf{k})$ for $c > 0$, whereas $\text{Var}(c\mathbf{k}) = c^2 \text{Var}(\mathbf{k})$ and carries units of degree squared. The mean degree is intensive, whereas $D = \bar{k}/(n-1)$ is not. The substitution therefore replaces two quantities that depend on the scale of the degree sequence and on the order of the graph by two that depend only on the shape of that sequence. This is the general reason; Proposition 7 is the case in which it can be checked exactly. Numerically, on Watts-Strogatz graphs with $K = 6$, $p = 0.1$ and n from 200 to 6400, the fitted slope of $\log_{10} \Omega$ against $\log_{10} n$ is -0.987 , consistent with Proposition 6, while that of $\tilde{\Omega}$ is $+0.009$. Both figures hold M fixed, as Proposition 7 does.

Remark 16 (The residual drift, and where it comes from). Holding M fixed is an idealisation, and on this family it is a false one: the algebraic connectivity of a Watts-Strogatz graph falls with its order, so $M = 1/\lambda_2(L)$ grows with n and neither index is exactly order-invariant. Measured over forty graphs per point at $K = 6$, $p = 0.1$ and n from 200 to 1600, with M evaluated rather than fixed, the fitted slopes become -1.160 for $\log_{10} \Omega$ and -0.160 for $\log_{10} \tilde{\Omega}$, against -1.009 and -0.010 for the same graphs with M held fixed. The whole of the difference is λ_2 , whose own slope is -0.151 on the same graphs; C , $\bar{k}$ and G_k are flat to within a few parts in a hundred across the range.

Two things follow. The substitution does most of what it is introduced to do, removing about seven eighths of the drift: a factor of roughly 14 per decade of n becomes a factor of roughly 1.4. And it does not remove all of it, because the concentration factor is not the only one that carries the order of the graph. Restoring exact comparability would require a modularity factor that is itself order-invariant, which we do not have and do not claim; the comparisons reported in this paper are at matched n for that reason.

Remark 17 (A third candidate for the concentration factor, with a dynamical rather than a dimensional justification). *The argument above justifies G_k on grounds of scale: it is invariant where $\text{Var}(\mathbf{k})$ is not. That argument says why the substitution restores comparability; it does not say why a concentration measure belongs in the denominator at all, which Remark 1 supplies only by ecological analogy. A third candidate carries both. Write*

$$\xi = \frac{\langle k^2 \rangle}{\langle k \rangle^2} = 1 + \frac{\text{Var}(\mathbf{k})}{\bar{k}^2} = 1 + \frac{\text{Coex}}{\bar{k}^2}, \quad (16)$$

the degree heterogeneity of the graph. It is Coex normalised by the square of the mean degree, hence dimensionless and scale-invariant like G_k ; and it is the quantity for which degree heterogeneity is shown to destabilise ecological networks with competitive and mutualistic interactions, entering a stability criterion of the form

$$\sigma \sqrt{(S-1)\mathcal{C}\xi} < d \quad (17)$$

for a community matrix with S species, self-regulation d and interaction strengths of dispersion σ [47]. We therefore record $\widehat{\Omega}_\xi = C\bar{k}/(M\xi)$ as a variant of Definition 4 on the same footing as the one stated there, and note that the two agree in direction while differing in how they weight the tail of the degree distribution.

Two cautions attach to importing (17), and both are about scope rather than about ξ . The first is notational and easy to misread: in the stability literature $\mathcal{C}$ denotes connectance, the fraction of realised links, which is the quantity written D here, whereas C in this paper is the clustering coefficient. Translated into the notation of Definition 1, the criterion reads $\sigma \sqrt{(n-1)D\xi} < d$; it says nothing about triadic structure. The second is substantive. In that criterion connectance and heterogeneity enter as a product, and increasing either is destabilising, whereas Definition 1 places D in the numerator and the concentration factor in the denominator, so the two are composed as a ratio and move Ω in opposite directions. This is the long-standing tension between the analytical stability criteria descending from [51, 52] and the empirical finding that connectivity accompanies function and robustness [45, 53, 48], and the two are not measuring the same notion of stability: the criterion concerns local asymptotic stability of a randomly parameterised community matrix, whereas the evidence behind the numerator of (3) concerns functional redundancy measured on ecosystem processes. We take from [47] the justification of the concentration factor and its normalisation, and not the composition of the index, and we state the difference rather than let the citation imply an endorsement it does not carry.

Remark 18 (Connectivity as a two-sided constraint, and what the index does not express). *Three independent lines converge on a statement the index as composed does not make. In the stability criteria discussed above, connectance enters as a destabilising quantity [51, 52, 47]. In communication graphs among language-model agents, reliability imposes a minimum degree and error propagation a maximum, so that the admissible graphs occupy an interval that may be empty [55]. And in the ecological literature from which this framework derives, the same shape has a name: Ulanowicz’s window of vitality, in which a flow network persists only in an intermediate band between too little organisation, which leaves it incoherent, and too much, which leaves it brittle for want of redundant pathways [56].*

Definition 1 is monotone in each of its factors, and in D in particular. It therefore expresses one edge of such a window and not the window itself: it says which direction is the resilient one, not where the direction reverses. This is a limitation of the composite and not of the underlying functional, since nothing in Propositions 3 or 4 depends on how the four factors are combined. It also identifies the natural extension: an index with an interior optimum in connectivity would require a second term penalising excess, of the kind that the overhead side of the ascendancy decomposition supplies. Appendix A carries that transplant out exactly for the setting of this paper, and reports both what it delivers, a closed form and a point prediction for the optimal

degree, and what it does not, since the resulting order parameter turns out to be a functional of the degree sequence alone and therefore blind to wiring. We record the convergence because the three sources share no method, and because agreement across epidemiology, random-matrix stability and information-theoretic ecology is the kind of evidence a purely structural argument cannot supply on its own.

One consistency check belongs here, because it delimits the reading rather than weakening it. The four families of Proposition 2 are compared at matched order and matched edge count, hence at matched mean degree, and an information-theoretic order parameter computed on them separates them hardly at all. The comparison that Proposition 2 reports is therefore one of arrangement at fixed connectivity, which is the regime in which the index is well posed. The limitation stated above concerns comparisons made across levels of connectivity, and no result of this paper makes one.

Remark 19 (When the shape-normalised index is exact, and how to check it in advance). Proposition 7 replaces D by $\bar{k}$ on the grounds that the mean degree is intensive. That premise is not automatic on a growing graph. If the edge count scales with the order as $m \sim n^\delta$, then $\bar{k} = 2m/n \sim n^{\delta-1}$ and $D \sim n^{\delta-2}$, so the substitution is exact precisely when $\delta = 1$ and leaves a residual exponent of order $\delta - 1$ otherwise. Real growing graphs are observed to densify, with δ typically between 1 and 2 [57], so the residual is generally not zero: on the syntactic networks of Section 7 the two accumulation exponents give $\delta = \beta_E/\beta_V = 0.955/0.77 = 1.24$, hence a predicted residual of +0.24 per decade of order in $\bar{k}$, which is what is measured there.

The practical consequence is that $\tilde{\Omega}$ carries its own applicability condition, and one that can be checked before the index is applied and without any additional data: δ is the slope of $\log m$ against $\log n$ across the units to be compared, and both quantities are already needed to compute D . Where δ is close to one the substitution removes the order dependence almost entirely; where the graph densifies strongly the substitution over-corrects, replacing a factor that falls with n by one that rises with it, and the shape argument of Remark 20 should not be expected to hold. We state the condition as a prediction of the scaling argument; a systematic measurement across families with different δ is left open.

Remark 20 (What Proposition 7 does not cover). Watts-Strogatz graphs have homogeneous degrees. The proposition establishes invariance to the order of a given graph; it does not establish invariance under a change of sampling effort when the graph is estimated and its degree distribution is heavy-tailed, which is the situation measured in Section 7. The analytical and the empirical statements agree in direction and are not the same claim. Nor does $\tilde{\Omega}$ inherit the ecological reading of Coex as a degree variance: the Gini coefficient is a concentration measure of the same sequence, so the reading transfers as one of inequality rather than of variance.

4. Curvature Interpretation: Triangles and Ollivier-Ricci Bounds

Before formalising that connection it is worth naming the closest application of this curvature to the problem the framework addresses. Sandhu, Georgiou and Tannenbaum [21] compute Ollivier-Ricci curvature on financial correlation networks as an indicator of market fragility and systemic risk: the same construction and the same purpose as the financial embodiment of this family, reached from discrete geometry rather than from triadic structure. A subsequent analysis [22] argues that the indicator identifies the size and duration of a crisis accurately but does not lead it, that is that it is an ex-post descriptor rather than an early warning. The same curvature has since been applied to market instability with an explicitly network-geometric reading [28], and, outside finance, as the basis of a community detection method [29], which is a second use of the same local quantity for a global structural purpose. Both bear on what follows. The bound of Theorem 2 relates the same common-neighbour counts that the curvature reads, so the two are not independent measurements of a network; and the ex-post limitation is

a property of that reading which a triadic reading does not automatically escape. We note also that the relation between Ollivier-Ricci curvature and the spectrum of the normalised Laplacian is established [23], which places the curvature of this section in the same spectral neighbourhood as Proposition 3 rather than outside it.

The connection to differential geometry is formalised through the *Ollivier-Ricci curvature* [8], a discrete analogue of Riemannian sectional curvature for metric spaces:

$$\kappa(x, y) = 1 - \frac{W_1(\mu_x, \mu_y)}{d(x, y)}, \quad (18)$$

where W_1 is the Wasserstein-1 (earth-mover) distance between the probability measures μ_x, μ_y over neighbours of x and y , and $d(x, y)$ is the graph distance.

Proposition 8 (ORC and Triangle Density). *For a graph G with adjacency matrix A and uniform random-walk measures $\mu_x(y) = A_{xy}/\deg(x)$, the Ollivier-Ricci curvature of an edge (x, y) is controlled by the number of common neighbours $t_{xy} = |N(x) \cap N(y)|$, which is exactly the contribution of that edge to $\text{Tr}(A^3) = \sum_{(x,y) \in E} 2t_{xy}$. The corresponding upper bound is Theorem 2; the condition under which the curvature is non-negative is Proposition 9.*

Theorem 2 (ORC Upper Bound). *Let $t_{xy} = |N(x) \cap N(y)|$ denote the number of common neighbours of x and y , and let $d_x = \deg(x)$, $d_y = \deg(y)$. Then:*

$$\kappa(x, y) \leq \frac{t_{xy}}{\min(d_x, d_y)}. \quad (19)$$

Proof. The optimal coupling between μ_x and μ_y matches each common neighbour $z \in N(x) \cap N(y)$ at zero transport cost. The minimum coupling assigns at most $t_{xy}/\min(d_x, d_y)$ of the total mass at zero cost, so $W_1(\mu_x, \mu_y) \geq 1 - t_{xy}/\min(d_x, d_y)$, giving $\kappa(x, y) \leq t_{xy}/\min(d_x, d_y)$. $\square$

Remark 21 (On the Lower Bound). *It is natural to conjecture a matching lower bound $\kappa(x, y) \geq t_{xy}/\max(d_x, d_y) - |d_x - d_y|/\max(d_x, d_y)$. We report that it does not hold in general. Numerical experiments with exact Wasserstein-1 computation on Erdős-Rényi and Barabási-Albert networks show it fails in networks with sparse triangle structure: when $t_{xy} = 0$ on most edges, many non-common neighbours lie at geodesic distance > 1 , making the actual transport cost higher than the bound assumes. The correct lower bound in full generality requires bounding the geodesic distances between non-common neighbours, a condition that holds for triangle-rich networks (clustering coefficient $C > 0.3$) but fails for sparse random graphs. Establishing tight lower bounds for arbitrary graphs is left as an open problem.*

Proposition 9 (Non-negativity Condition for Lin-Lu-Yau Curvature). *Let $\{x, y\} \in E(G)$ with $d_x = \deg(x) < d_y = \deg(y)$, let $\Delta = N(x) \cap N(y)$ and $\delta = |\Delta|$, and write $A = N(x) \setminus N[y]$ and $B = N(y) \setminus N[x]$. By [18] (Theorem 1.2), if*

$$\delta \geq \frac{2d_x d_y - 2d_x - 2d_y}{d_y + 2d_x}, \quad (20)$$

then the Lin-Lu-Yau Ricci curvature [19] $\kappa^{\text{LLY}}(x, y) \geq 0$. Moreover, $\kappa^{\text{LLY}}(x, y) > 0$ unless simultaneously: (i) equality holds in (20), (ii) there are no edges between Δ and B , and (iii) $\text{dist}(u, w) = 3$ for all $u \in A$, $w \in B$.

Remark. Condition (20) makes non-negative curvature sufficient once the common neighbourhood is large relative to the degrees: the right-hand side grows like $2d_x/3$ when d_y is much the larger of the two, so the condition asks the edge to close a number of triangles comparable to the degree of its lower-degree endpoint. The hypothesis $d_x < d_y$ is strict in [18], so the statement does not specialise to regular graphs, for which the exact formulas of that work (Theorem 1.1) apply instead.

The general bounds on $\kappa^\alpha(x, y)$ for all $\alpha \in [0, 1]$ and arbitrary degree sequences are given by Theorem 5.3 of [18], which extends earlier results of Jost-Liu [24] to non-regular graphs.

Connection to $\text{Tr}(A^3)$. The condition (20) depends on $\delta = t_{xy}$, the same common-neighbour count that determines the local contribution to $\text{Tr}(A^3) = \sum_{(x,y) \in E} 2t_{xy}$, and it points the same way as Theorem 2: triangle-rich edges are the ones whose curvature is non-negative. Edges satisfying (20) contribute non-negatively to the total Ollivier-Ricci curvature, while edges with $\delta < \frac{2d_x d_y - 2d_x - 2d_y}{d_y + 2d_x}$ are the ones that may have $\kappa < 0$ and act as topological bridges. Those are the edges that close few triangles and therefore contribute little to $\text{Tr}(A^3)$, so the curvature reading and the triadic one agree on which edges are which. What they do not agree on is the direction of the composite: a triangle-poor bridge is a negative-curvature edge, and it is the distribution of the triangles across nodes, read by Coex rather than by $\text{Tr}(A^3)$ alone, that carries the monopoly signature (Remark 10).

Corollary 2 (Threshold for Positive Total Curvature). *The upper bound $\Sigma\kappa^+ \leq \text{Tr}(A^3)/(2d_{\min})$ holds on every network in Table 3. It is an inequality and not an estimate: edges satisfying (20) contribute non-negatively to $\Sigma\kappa^+$, whereas edges with $\kappa < 0$ contribute nothing to it while their triangles still enter $\text{Tr}(A^3)$. The bound is therefore expected to be loose in graphs where a large fraction of edges act as bridges, and looser still when $d_{\min}$ is small; see Remark 22.*

Consequence for $\text{Tr}(A^3)$. Since $\text{Tr}(A^3) = 6N_\Delta = \sum_{(x,y) \in E} 2t_{xy}$, the upper bound gives:

$$\sum_{(x,y) \in E} \kappa^+(x, y) \leq \frac{\text{Tr}(A^3)}{2d_{\min}}. \quad (21)$$

Empirical validation. Using exact W_1 computation via optimal transport (Wasserstein-1 distance between neighbour distributions), we validate on six networks. Results in Table 3 show that all six satisfy the upper bound (21). The bound is, however, far from tight, and its slack does not order by clustering: the ratio $\Sigma\kappa^+/\text{UB}$ is 0.065 for Karate Club, essentially the same as 0.047 for the Erdős-Rényi graph and 0.069 for the Barabási-Albert graph, even though the first has a dense triangle structure and the other two do not. Remark 22 identifies what does control the slack. The companion lower bound $\text{Tr}(A^3)/(2d_{\max})$, which is natural to conjecture alongside (21), is not satisfied on the two sparse random graphs: it evaluates to 9.00 against $\Sigma\kappa^+ = 1.90$ for Erdős-Rényi and to 7.83 against 3.25 for Barabási-Albert. This is the aggregate counterpart of the edge-wise failure reported in Remark 21, and it is why only the upper bound is claimed here.

Table 3: ORC upper bound validation. $\Sigma\kappa^+$ is computed with exact Wasserstein-1 optimal transport; $\text{LB} = \text{Tr}(A^3)/(2d_{\max})$ is the lower bound one would naturally conjecture alongside it, and $\text{UB} = \text{Tr}(A^3)/(2d_{\min})$ is the bound of (21). All six networks satisfy the upper bound; the conjectured lower bound fails on the two sparse random graphs. Slack is governed by $d_{\min}$, not by clustering; see Remark 22. The values were computed with `colabs/fsri_colab4b_orc_exact.py`, which is included in the accompanying repository.

Network	N	M	$d_{\min}$	$d_{\max}$	LB	$\Sigma\kappa^+$	UB
Karate Club [16]	34	78	1	17	7.94	8.74	135.00
Florentine Families [17]	15	20	1	6	1.50	1.50	9.00
Les Misérables	77	254	1	36	38.92	82.40	1401.00
Davis Southern Women	18	139	11	17	111.35	111.78	172.09
Erdős-Rényi ($n=40$)	40	110	2	9	9.00	1.90	40.50
Barabási-Albert ($n=40$)	40	111	3	18	7.83	3.25	47.00

Remark 22 (Why the Bound Is Loose). *The slack in (21) is governed by $d_{\min}$ rather than by clustering. A single pendant vertex sets $d_{\min} = 1$ and doubles the bound relative to an otherwise identical graph with $d_{\min} = 2$, however many triangles the graph contains. Three of the six networks in Table 3 have $d_{\min} = 1$ (Karate Club, Florentine Families, Les Misérables), and for these the reported UB is exactly $\text{Tr}(A^3)/2$: $135.00 = 270/2$, $9.00 = 18/2$ and $1401.00 = 2802/2$, for 45, 3 and 467 triangles respectively. This is why Karate Club and Les Misérables, both triangle-rich, show slack indistinguishable from that of the sparse random graphs, while Davis Southern Women, the only network here with $d_{\min} = 11$, is the one case where the bound is nearly attained (111.78 against 172.09). A second and independent effect acts in the sparse random graphs, where most edges have $\kappa < 0$ and act as bridges rather than closing triangles, so that $\Sigma\kappa^+$ is small in absolute terms as well. Neither effect is evidence against the bound, and neither is captured by the clustering coefficient.*

This gives $\mathcal{T}(W)$ a precise geometric meaning: it measures the *total positive curvature* of the weight graph. Minimising $\mathcal{T}(W)$ flattens the curvature distribution, moving the network toward a configuration where curvature is uniformly distributed, the network analogue of a maximally symmetric Riemannian manifold.

Remark 23 (Continuum limits and other curvature observables: what is not claimed here). *Two results in the literature extend the discrete construction toward continuous geometry, and we record them as context rather than as contributions of our own. The Ollivier-Ricci curvature of random geometric graphs embedded in a Riemannian manifold converges in the continuum limit to the Ricci curvature of that manifold, and the result extends to Lorentzian signature [30]; building on it, the construction has been defined for causal sets using Lorentzian optimal transport [31]. Where a network encodes causal relations in that sense [32], its curvature approximates a Ricci scalar, and one may read the two ends of Ω as regions where geodesics focus and defocus. We derive nothing in that direction, and a reader who discounts this remark entirely loses nothing that the rest of the paper relies on. Separately, nonperturbative quantum gravity has its own curvature observable for discrete geometries [35, 36, 37], which is a different construction: it compares distances between spheres rather than transporting neighbour measures, and admits no triangle-counting identity, so the bound above does not transfer to it. Whether $\mathcal{T}$ has an analogue for quantum channels under non-commutative transport metrics [58] is open.*

5. Application: Omega-S Neural Network Regulariser

5.1. From Network Theory to Neural Networks

The FSRI framework has an immediate application to neural network regularisation. For a weight matrix $W^{(\ell)} \in \mathbb{R}^{m \times n}$ in layer ℓ of a neural network, we define the layer-wise topological pressure:

$$\mathcal{T}_\ell = \frac{\lambda_\Omega}{|\text{numel}(W^{(\ell)})|} \cdot \widehat{\text{Tr}}(W^{(\ell)} W^{(\ell)\top})^3, \quad (22)$$

and the total regularisation penalty:

$$\mathcal{L}_\mathcal{T} = \sum_\ell \mathcal{T}_\ell. \quad (23)$$

The Omega-S regulariser adds $\mathcal{L}_\mathcal{T}$ to the task loss every K training steps. By Proposition 5, this drives the network’s weight matrices toward minimum-entropy-production configurations, that is, states where structural monopolies are suppressed.

One point should be stated plainly, because the shared name invites the opposite reading. *The regulariser minimises $\mathcal{T}(W)$ and never computes Ω .* No clustering coefficient, connection density, modularity or degree variance enters the training loop: the penalty is the Hutchinson estimate of $\text{Tr}(WW^\top)^3$ and nothing else. The index and the penalty are related by interpretation, through Definition 3 and Proposition 5, and not by computation. This is why the tension

identified in Remark 10 does not arise in the applied setting: what is minimised is the topological pressure, whose thermodynamic reading is unambiguous, rather than the composite index, whose response to that minimisation depends on how Coex moves alongside it and is therefore a measured outcome.

5.2. Empirical Validation

The Omega-S regulariser [1] validates the theoretical predictions:

Table 4: Omega-S empirical results, with the setting for each row. All values are as reported in the applied paper [1]. Retention is measured on Llama-3-8B (LoRA, code $\rightarrow$ prose) as HumanEval pass@1 post-prose / post-code; the degree-variance row is measured on the three-layer MLP with Split-MNIST and the structured-sparsity row on GPT-2, not on Llama-3-8B.

Result	Value	Interpretation
Code capability kept	0.173 $\rightarrow$ 0.238, 9/10	Primary metric, vs no regulariser; sign test $p = 0.011$
Same, as retention ratio	62.9% $\rightarrow$ 84.1%	The same comparison, in ratio form
Retention vs weight decay	61.9% $\rightarrow$ 84.1%, 10/10	Secondary, $p = 0.002$; thin selection
Retention vs EWC	69.1% $\rightarrow$ 84.1%, 8/10	Secondary, $p = 0.014$; thin selection
vs row-norm control (own sweep)	8/10, $p = 0.055$	Not reducible to norm balancing
Degree variance reduction (MLP)	41.94 $\rightarrow$ 0.136	8 $\times$ the reduction obtained by weight decay (1.12)
Structured sparsity (GPT-2)	-0.72% vs -0.77%	No gain over group lasso alone; the compression is group lasso’s
Training overhead	$< 4\%$ per step	Analytic, penalty every 10 steps
Latency overhead (2-GPU FSDP)	$+1.5\%$	Measured; zero inter-GPU cost

The retention result is particularly significant from the thermodynamic perspective: minimum-entropy-production states (low $\mathcal{T}$) are more stable under perturbation (new training data), consistent with Prigogine’s characterisation of stationary non-equilibrium states as maximally robust against small deviations.

6. Applications Beyond Neural Networks

The FSRI framework is domain-agnostic, and the readings below are offered as interpretations of the same functional in other settings rather than as results. None of them is measured here.

Ecological networks. A network dominated by a single node (Coex $\rightarrow \infty$, $C \rightarrow 0$) collapses $\Omega \rightarrow 0$, which the framework reads as systemic fragility; the Florentine Medici marriage and credit network [17] is the standard illustration of that topology. Soil microbial networks under resource abundance exhibit the same monopoly structure [14], and land-use intensification is reported to reduce network complexity and the abundance of keystone taxa [54]. The one experimental test of the direction this reading assumes manipulates soil biodiversity in grassland microcosms and finds that network complexity predicts ecosystem multifunctionality, with the relation surviving a correction for richness [45]; that result is about complexity and function rather than about Ω , and we cite it as the motivation for the numerator of (3) and not as a validation of the composite.

Financial systems. The Minsky Moment [15], the sudden collapse of a leveraged financial system, is a structural phase transition from high- Ω (distributed, mutualistic) to low- Ω (monopolistic, fragile) topology under energy abundance ($E_{\text{ext}} = \text{credit availability}$).

Power grids. Cascade failure in a linear (high- M , low- C) grid reads in the same terms, as a topological phase transition triggered by a central shock.

Medical systems. Microbiome dysbiosis is sometimes described in the same terms, with energy abundance driving specialisation (M increases, C decreases, Coex increases, Ω collapses). We flag this reading as the least supported of those listed here, and as an illustration rather than an expectation. Evidence from the human gut does not line up with it uniformly: co-occurrence networks in inflammatory bowel disease are reported to be reorganised rather than simply degraded, and the healthy network is the one more easily disrupted by removal of a few key taxa [50]. The orientation of M in this setting is subject to Remark 3. The reading is also subject to the sampling condition of Section 7: the node set of a co-occurrence network of taxa grows with sequencing depth, so readings taken at unequal depth are not comparable without an explicit standardisation, and the factor most affected, Coex, is the one this reading relies on.

Multi-agent language-model systems. A recent analysis of communication graphs among language-model agents derives a two-sided constraint on connectivity of exactly the shape this framework describes, and reaches it from epidemiology rather than from triadic structure [55]. A clean-task reliability target imposes a minimum degree, because a node needs enough independent opinions for a majority vote to reach the required accuracy; an error-propagation target imposes a maximum, because at fixed exposure per edge the invasion factor grows with the spectral radius of the adjacency matrix. The feasible regular degrees are then confined to an interval, and the authors characterise when that interval is empty. The construction differs from ours in every detail, and we record the structural agreement rather than any numerical one; its relation to Definition 1 is the subject of Remark 18.

7. Limitations and Future Work

We set out the scope of each result, distinguishing what is established from what remains conjectural.

Scope of the ORC-triangle connection. Proposition 8 is proved using Jost-Liu [24] for regular and semi-regular graphs. The approximation $\kappa^+(x, y) \propto |N(x) \cap N(y)|/\bar{d}$ fails for highly heterogeneous degree distributions; a fully general proof requires the exact Wasserstein-1 computation and is left for future work.

Scope of the FSRI proportionality. Proposition 6 derives the exact FSRI formula for Watts-Strogatz networks, $\Omega = [(3K - 6)/(4K - 2)] \cdot (1 - p)^2/(np)$, which shows that the proportionality $\Omega \propto (S_{\text{env}}/E_{\text{ext}})^2$ does not hold in general. Substituting $\text{Var}(k) = Kp(1 - p)$ and $E_{\text{ext}} = K$ into (13) gives $(S_{\text{env}}/E_{\text{ext}})^2 = p(1 - p)/K$ and hence the exact identity

$$\Omega = \frac{1}{nK} \cdot \frac{3K - 6}{4K - 2} \cdot \frac{(1 - p)^3}{(S_{\text{env}}/E_{\text{ext}})^2},$$

in which the ratio appears in the *denominator*. The relation is inverse rather than proportional, which is what produces the negative correlations $r = -0.407$ to -0.573 reported above rather than merely weak positive ones. Deriving the corresponding expression for other network models (Barabási-Albert, Erdős-Rényi) remains open.

Sampled graphs and the growing node set. Remark 5 states the condition under which two values of Ω may be compared: matched order and matched density. A stronger condition applies when A is not given but estimated from a finite sample whose node set grows with the sampling effort. Co-occurrence networks of microbial taxa at different sequencing depths, and lexical or syntactic networks at different corpus sizes, are of this kind: additional sampling adds vertices rather than only refining the weights among those already present. In ecological terms this is the sampling regime of a metacommunity with a non-zero speciation rate [43], in which the local sample is drawn from a regional pool that is itself still being added to, so that richness does not saturate and neither does any factor computed from it. Matching the sample size is then not sufficient, because two units of equal sample size sit at different points of their respective accumulation curves.

We measured the effect on syntactic dependency networks built from Universal Dependencies treebanks for 28 languages, at token budgets from 2 000 to 32 000, fitting $\log_{10}(\cdot) = a + b \log_{10} N$ per language. The accumulation exponents are $\beta_V = 0.77 \pm 0.06$ for the vertex set and $\beta_E = 0.955 \pm 0.017$ for the edge set. The former is the sublinear vocabulary growth of Herdan and Heaps [41], measured here at the small end of the corpus range where that exponent is known to exceed the 0.4 to 0.6 typical of large corpora. The latter says that distinct dependency pairs accumulate almost linearly with the sample: the edge set is nowhere near saturation in this range, and the tightness of β_E across typologically distant families is worth noting in itself.

All four factors drift, and the difference between them is one of degree rather than of kind: $b = +1.03$ for Coex and -0.59 for D , against -0.14 for C and -0.09 for M . For the index itself $b = -1.68$, so a tenfold increase in corpus size divides Ω by roughly fifty. Two of these slopes carry different weight, and we separate them rather than report them alike. That of D follows algebraically from β_V and β_E , since D is a fixed function of n and m , and is therefore a consistency check on the fits and not independent evidence. That of Coex does not, since it depends on the shape of the degree distribution and not only on the two accumulation rates, and it is the slope that carries information. The drift is not an artefact of the small end of the range: the slope of Coex changes by -0.02 between the lower and upper halves of the budgets tested. Neither does it extrapolate, since β_V is itself known to fall as corpora grow [42].

Two routes are available in the literature. The first is to standardise by sample *coverage* rather than by sample size, following Chao and Jost [39], who show that equal-sized samples systematically bias comparisons because a size sufficient to characterise a poorer community is insufficient for a richer one, and that coverage is estimable from the singleton count. The second is to replace the concentration factor by a parameter of the generating process rather than a statistic of the realised sample, for instance the fitted tail exponent γ of the degree distribution [40]. On these corpora that exponent is estimable and lies below three throughout ($\gamma = 2.67 \pm 0.22$ over 582 samples, range 1.92 to 2.99), which makes the divergence condition of Remark 5 concrete rather than hypothetical. It does not account for the observed rate: the standard scaling for a tail exponent of that size predicts a drift about an order of magnitude smaller than the one measured. It also changes what the factor means, breaking the reading of Coex as a degree variance on which the competitive-exclusion analogy of Remark 1 rests.

The same two factors identified analytically in Proposition 7 are the ones that carry the drift here, and the shape-normalised index of Definition 4 largely removes it. The Gini coefficient of the degree sequence drifts at $b = +0.09$ and the normalised entropy of that sequence at $b = -0.02$, against $+1.03$ for $\text{Var}(\mathbf{k})$. Substituting the concentration factor alone leaves the index at $b = -0.73$, because D still drifts at -0.59 and, as noted above, the slope of D is $\beta_E - 2\beta_V$, a property of the sampling regime rather than of the statistic chosen. Replacing D by the mean degree as well, whose slope is $\beta_E - \beta_V = 0.19$, gives $b = +0.04$ for $\tilde{\Omega}$, a factor of roughly 1.1 per decade of corpus size against the factor of 47 carried by Ω as defined. Restoring comparability therefore requires changing two factors rather than one, and the pair that works empirically is the pair the proposition identifies.

We none the less keep Ω as the index of Definition 1 and $\tilde{\Omega}$ as a variant for sampled graphs, for the reason given in Remark 20: the proposition covers graphs with homogeneous degrees, the measurement covers estimated graphs with heavy-tailed ones, and the second does not follow from the first. What can be said is that the variant is not an ad hoc repair. It has the same analytical status on the Watts-Strogatz family as the index it modifies, and it is the substitution that the scale argument predicts in advance.

Two elements of the treatment above have published precedents that are worth naming, because both are sometimes taken to be idiosyncratic. The first is the construction: inferring a single association network over all units and then reading, for each unit, the subnetwork induced by the taxa it contains is the procedure used to relate microbiome complexity to ecosystem multifunctionality in grassland microcosms [45], and a per-sample connectivity statistic computed

from a global matrix of pairwise associations is the basis of the cohesion metric of [46]. The second is the correction: the same work observes that network complexity depends on the number of nodes, detrends for richness before interpreting, and reports that the residual relation to function survives that detrending [45]. The condition stated here is therefore not a new objection to an established practice but the quantitative form of a correction that practice already applies.

A limitation applies to the measurement itself. Universal Dependencies provides several treebanks of differing genre for some languages, and the dispersion of these quantities between treebanks of one language is not small relative to the dispersion between languages. Measured on sixteen treebanks across five languages at a matched budget, the ratio of the second to the first is 3.6 for the Gini coefficient, 3.2 for the normalised entropy and 2.15 for $\text{Var}(\mathbf{k})$. Comparison of network quantities estimated from treebanks is therefore genre-contaminated across languages as well as sampling-dependent within them, and the margins above should be read accordingly.

Status of the ORC numerical validation. The Colab 3 experiment confirms numerically that ORC $\kappa(x, y)$ correlates strongly with edge contribution to $\text{Tr}(A^3)$ in Watts-Strogatz networks, supporting Proposition 8 empirically. The evidence is numerical rather than analytical, and restricted to that family.

Scope of the spectral identity. Proposition 3 constrains what can be claimed for the global index: as a scalar detector, $\mathcal{T}$ should be benchmarked against, and is expected to tie, other spectral functionals. Claims of superiority over spectral methods are defensible only for the node-wise field (Corollary 1) or for the training-time gradient flow, not for the global statistic. The signed extension is stated but not yet validated empirically; whether signed $\mathcal{T}$ improves detection in domains where sign is meaningful is an open question.

The geometric reading is limited to what Section 4 establishes, namely the inequality relating common-neighbour counts to $\text{Tr}(A^3)$ and its measured slack. Continuum limits, causal sets and the curvature observables of nonperturbative quantum gravity are noted in Remark 23 as context; we derive no result in that direction and none of our claims depends on one.

8. Conclusion

We have presented a mathematical framework connecting the Functional Symbiotic Resilience Index Ω to non-equilibrium thermodynamics via the topological pressure functional $\mathcal{T}(W) = \text{Tr}(WW^\top)^3$.

The key results are:

1. $\mathcal{T}(W)$ is proportional to the global clustering coefficient of the induced weight graph, providing a precise topological interpretation.
2. The gradient $3(WW^\top)^2$ is the *topological pressure* on each weight, whose Gini coefficient discriminates monopoly networks ($G = 0.986$) from uniform networks ($G = 0.272$) by $3.6\times$.
3. The gradient flow $\dot{W} = -\lambda\nabla_W\mathcal{T}(W)$ implements Prigogine’s minimum-entropy-production principle in the space of weight configurations.
4. Phase transitions in triangle density (fugacity mechanism) confirm the thermodynamic analogy: $\text{Tr}(A^3)$ increases $76\times$ over the transition, while the Ollivier-Ricci curvature interpretation provides a geometric framework consistent with the relational motivation of the index.
5. The index is not invariant to the order of the graph, and the dependence is carried by exactly two of its four factors. Replacing them by their shape-normalised counterparts, the mean degree and the Gini coefficient of the degree sequence, removes the dependence

exactly on Watts-Strogatz graphs (Proposition 7) and reduces it from a factor of 47 to a factor of 1.1 per decade of sampling effort on estimated graphs (Section 7).

Since several of the results above are restrictive, we state in one place what the framework claims after them. Three things survive. The node-wise field $\text{diag}(A^3)$ is not a function of the spectrum (Proposition 4) and is therefore the component that no eigenvalue-based scalar can reproduce. The training-time gradient flow of Proposition 5 is a construction rather than a statistic, and its empirical outcome is reported in Section 5. And the global scalar Ω remains a usable descriptive index within the conditions of use set out in Remark 5 and Section 7, namely at matched graph order and density and, for graphs estimated from samples, under an explicit standardisation, or in the shape-normalised form of Definition 4 where those conditions cannot be met. What we do not claim is that the global scalar dominates other spectral functionals as a detector; Proposition 3 forbids it, and we say so rather than leave it to be found.

The Omega-S neural network regulariser is a concrete instantiation of the framework rather than a proof of it. Minimising $\mathcal{T}(W)$ during training yields networks that are more stable under sequential learning (mean retention 61.9% $\rightarrow$ 84.1% over tuned weight decay across ten seeds, ten of ten positive) and, in combination with group lasso, structurally prunable, although the compression obtained in that setting is attributable to the group-lasso term and the marginal contribution of the penalty remains to be isolated. These are empirical outcomes consistent with the minimum-entropy-production reading developed above; they do not establish that reading, which would require measuring entropy production directly rather than inferring it from the structural proxy. We record the distinction because the thermodynamic interpretation is the more speculative part of this work, and the empirical results stand independently of whether it is ultimately upheld.

Declaration of LLM Usage

Large language models were used as writing and coding assistants during the preparation of this manuscript: to improve language clarity and presentation, to assist with coding and debugging for the numerical verifications, and to help identify related work. All technical ideas, theoretical results, proofs, experimental design and final interpretations were developed, verified and approved by the author.

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A. The Two-Sided Reading Made Exact: Ascendency on an Unweighted Graph

Remark 18 records that Definition 1 is monotone in each of its factors and therefore expresses one edge of a two-sided constraint rather than the constraint itself, and names Ulanowicz’s window of vitality [56] as the ecological antecedent. That framework is stated for weighted flow networks and its transplant to the setting of this paper is not obvious, so we carry it out here. The result is short, exact, and it settles what such a transplant can and cannot deliver.

A.1. Setup

Let A be the adjacency matrix of an undirected unweighted graph with m edges and degree sequence $\mathbf{k} = A\mathbf{1}$. Ulanowicz's decomposition takes a matrix of flows; on an unweighted graph the only choice that introduces no free parameter is the uniform one,

$$p_{ij} = \frac{A_{ij}}{2m}, \quad p_{i\cdot} = p_{\cdot i} = \frac{k_i}{2m}. \quad (24)$$

The capacity is $H = -\sum_{ij} p_{ij} \log p_{ij}$, the ascendancy density is the average mutual information $\mathcal{A} = \sum_{ij} p_{ij} \log(p_{ij}/(p_{i\cdot}p_{\cdot j}))$, the overhead is $\Phi = H - \mathcal{A}$, and the degree of order is $\alpha = \mathcal{A}/H$.

A.2. Three exact statements

Lemma 1 (Capacity is fixed by the edge count). *Under (24), $H = \log(2m)$ for every graph.*

Proof. There are exactly $2m$ non-zero entries of p , each equal to $1/2m$, so

$$H = -2m \cdot \frac{1}{2m} \log \frac{1}{2m} = \log(2m). \quad \square$$

Proposition 10 (Ascendancy is mean surprisal against the configuration null). *Let $P_{ij}^0 = k_i k_j / 2m$ be the expected number of edges between i and j under the configuration model. Then under (24)*

$$\mathcal{A} = \frac{1}{2m} \sum_{(i,j): A_{ij}=1} \log \frac{2m}{k_i k_j} = \langle -\log P_{ij}^0 \rangle_{A_{ij}=1}, \quad (25)$$

the mean Shannon surprisal of the realised edges under that null, and equivalently the Kullback-Leibler divergence of the edge distribution from the product of its marginals.

Proof. Substituting (24) into the definition of $\mathcal{A}$, each non-zero term contributes

$$\frac{1}{2m} \log \left[\frac{1/2m}{k_i k_j / (2m)^2} \right] = \frac{1}{2m} \log \left(\frac{2m}{k_i k_j} \right),$$

and $2m/(k_i k_j) = 1/P_{ij}^0$. $\square$

This identifies the construction rather than merely permitting it. The null P^0 is the same one against which Newman modularity is defined; the two quantities are different divergences from it, modularity taking the difference $A_{ij} - P_{ij}^0$ and ascendancy the logarithm of the ratio. The transplant is therefore not an analogy imported from another field but a second reading of an object already standard in network science.

Corollary 3 (Closed form). *Write $\langle \log k \rangle_w = \sum_i (k_i/2m) \log k_i$ for the degree-weighted mean of $\log k$. Then*

$$\Phi = 2\langle \log k \rangle_w, \quad \mathcal{A} = \log(2m) - 2\langle \log k \rangle_w, \quad \alpha = 1 - \frac{2\langle \log k \rangle_w}{\log(2m)}. \quad (26)$$

Proof. Expand (25): $\sum_{ij} A_{ij} \log k_i = \sum_i k_i \log k_i$ and likewise for j , so the double sum contributes $2 \sum_i k_i \log k_i = 4m \langle \log k \rangle_w$. Divide by $2m$ and subtract from $\log(2m)$; then use Lemma 1. $\square$

Corollary 4 (Optimal degree on regular graphs). *For a k -regular graph on n vertices, $\mathcal{A} = \log(n/k)$, $\Phi = 2 \log k$ and $\alpha = \log(n/k)/\log(nk)$. The robustness function $R(\alpha) = -\alpha \log \alpha$ of [56] is maximal at $\alpha = 1/e$, which holds exactly when*

$$\frac{\log k^*}{\log n} = \frac{e-1}{e+1} = 0.4621\dots, \quad \text{that is} \quad k^* = n^{(e-1)/(e+1)}. \quad (27)$$

Two features of (27) are worth stating. The optimal degree grows sublinearly with the order, so the optimal density $D^* = n^{-2/(e+1)}$ tends to zero: the two-sided reading prescribes increasing sparsity with size, which is the regime in which large empirical networks are found. And (27) is a point prediction, falsifiable on any family of graphs with varying order, which is more than the monotone reading of Definition 1 can offer.

A.3. What the transplant cannot do

Remark 24 (The order parameter is a degree-sequence statistic). *Equation (26) depends on the graph only through m and $\mathbf{k}$. Hence $\mathcal{A}$, Φ , α and R are invariant under any rewiring that preserves the degree sequence, and in particular they cannot distinguish a clustered graph from its configuration-model randomisation. A numerical check on double-edge swaps confirms this: the global clustering coefficient moves while α is unchanged to six decimal places.*

This bounds the reading of Remark 18 more precisely than that remark could. The window obtained by this route is a statement about degree distributions and not about wiring. It also corrects a natural but wrong way to describe the pairing: α and Coex are not a level coordinate and a shape coordinate drawn from different parts of the graph, since both are functionals of $\mathbf{k}$ alone, differing in weighting, logarithmic against quadratic, with Coex a pure dispersion that vanishes on regular graphs and α moving with the level as well. The genuine wiring content of Definition 1 is carried by C and M , and no quantity in this appendix touches it. A three-way reading is therefore the accurate one: the degree sequence, read by α or by Coex ; the wiring at fixed degrees, read by C and M ; and the order.

Two further scope conditions apply. The uniform flow (24) is a choice, and although it is the only parameter-free one available on an unweighted graph, the quantities above are not the ascendancy of a measured flow network. And Ulanowicz’s overhead decomposes into imports, exports, dissipation and redundancy on a directed flow network, of which a symmetric A retains only the last; Φ above should be read as a redundancy and not as the full overhead.